

Mini-Review

Fields - Idea - understand/generalize $\mathbb{R}$

- Must have commutative, associative addition w/ add. identity 0 & additive inverses
- Must have commutative, associative, distributive mult. w/ identity 1 & mult. inverses for non-zero elements

Major Examples: $\mathbb{Q}, \mathbb{R}, \mathbb{C}$, & $\mathbb{Z}_2 = \{0, 1\}$ defined

$$\text{by } 0+1=0 \quad (\text{i.e. } 1=-1). \\ 1+1=0$$

Vector Spaces - Idea - understand/generalize $\mathbb{R}^2, \mathbb{R}^3, \dots$

- Some base field $\mathbb{F}$
 - $\rightarrow$ Commutative, associative addition $+$: $V \times V \rightarrow V$ w/ identity 0 & add. inverses
- $\rightarrow$ Scalar mult. $\cdot$: $\mathbb{F} \times V \rightarrow V$ that is associative, distributive, & has $1 \cdot v = v \quad \forall v \in V$.

Major Examples: $\mathbb{F}^n, \mathbb{F}^S = \{f: S \rightarrow \mathbb{F}\},$

$$\mathcal{P} = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in \mathbb{F} \right\}$$

Subspaces - Idea - V.S. in another

- $A \subseteq V$, V a v.s. over $\mathbb{F}$, & A a v.s. under inherited op.s.
- Criterion: $[\forall v, w \in A \text{ & } c \in \mathbb{F}, \quad cv + w \in A] \Rightarrow \text{Subspace}$
- $A+B = \{a+b \mid a \in A, b \in B\}$ also a Subspace.

$\rightarrow$ A direct sum $A \oplus B$ if $A \cap B = \{0\}$

$\Leftrightarrow$ all $v \in A+B$ may be written uniquely as $v = a+b$.

Solutions

Worksheet 2: September 4th

Problem 1. Prove that the sum of the first n positive odd integers is n^2 .

Induction $1 = 1^2$ gives $n=1$
 Assuming n , $\sum_{i=0}^n 2i+1 = 2n+1 + \sum_{i=0}^{n-1} 2i+1$
 $= 2n+1 + n^2$
 $= (n+1)^2$

Problem 2. Define $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$.

(a) For a and b not both 0, rationalize

$$\frac{1}{a + b\sqrt{2}}$$

to show that this is an element of $\mathbb{Q}[\sqrt{2}]$.

$$\frac{1}{a + b\sqrt{2}} = \frac{a}{a^2 - 2b^2} - \frac{b}{a^2 - 2b^2} \sqrt{2} \in \mathbb{Q}$$

(b) Prove, using the field axioms, that $\mathbb{Q}[\sqrt{2}]$ is a field.

~~Additional properties are from $\mathbb{R}$.~~

Fastest - $\mathbb{Q}[\sqrt{2}] \subseteq \mathbb{R}$ is closed under addition & multiplication

Problem 3. Let $M(\mathbb{R}) = \{(a_1, a_2, a_3) \mid a_i \in \mathbb{R}\}$ for .

(a) Define $(a_1, a_2, a_3) + (b_1, b_2, b_3) = (a_1 + b_1, a_2 - b_2, b_3 - a_3)$ and $c * (a_1, a_2, a_3) = (ca_1, ca_2, ca_3)$. Do these operations make $M(\mathbb{R})$ a vector space?

No: Addition isn't commutative.

(b) For $\mathbb{F}_2 = \mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$, define $M(\mathbb{F}_2) = \{(a_1, a_2, a_3) \mid a_i \in \mathbb{F}_2\}$. With the operations defined above, is this a vector space?

Yes: $-b = b$ in $\mathbb{F}_2$, so $(a_1, a_2, a_3) + (b_1, b_2, b_3) = (a_1 + b_1, a_2 + b_2, a_3 + b_3)$

Problem 4. Let V be a vector space and A, B, C be subspaces of V such that $A, B \subset C$. Show that $A + B \subset C$ (so $A + B$ is the smallest subspace containing A and B).

$$\text{For } a \in A, b \in B, a, b \in C \Rightarrow a+b \in C \Rightarrow A+B \subset C.$$

Problem 5. Let $P = \{a_0 + a_1x + \dots + a_nx^n \mid n \text{ finite}, a_i \in \mathbb{R}\}$ be the space of real polynomials. Using the subspace criterion, show that the space $E \subset P$ of polynomials with only even powers of x and the space $O \subset P$ of polynomials with only odd powers of x are subspaces. Show further that $P = E + O$.

$$\rightarrow \text{For } p(x) = \sum_{i=0}^n a_i x^{2i}, c \in \mathbb{R}, cp(x) = \sum c a_i x^{2i} \\ \text{still has only even powers. For } q(x) = \sum_{j=0}^m b_j x^{2j+1}$$

$$cp(x) + q(x) = \sum_{i=0}^{\max(m,n)} (ca_i + b_i) x^{2i} \text{ is still in } E.$$

Hence, E is a subspace.

$$\rightarrow \text{For } p(x) = \sum_{i=0}^n a_i x^i, p(x) = \sum_{i=0}^{\lfloor n/2 \rfloor} a_{2i} x^{2i} + \sum_{j=0}^{\lfloor \frac{n-1}{2} \rfloor} a_{2j+1} x^{2j+1} \\ \in E + O.$$

Problem 6. Let $U = \{(x, z-x, z) \mid x, z \in \mathbb{R}\} \subset \mathbb{R}^3$. Find V so $\mathbb{R}^3 = U \oplus V$.

$$\rightarrow \text{For } V = \{(0, y, 0) \mid y \in \mathbb{R}\}, V \cap U = \{0\} \text{ and given } (x, y, z),$$

$$(x, y, z) = (x, z-x, z) + (0, y+x-z, 0)$$

$$\text{Shows } \mathbb{R}^3 \subset U+V. \text{ Hence, } U \oplus V = \mathbb{R}^3.$$

Problem 7. Show that $\text{span}\{v_1, \dots, v_n\} = \text{span}\{v_1\} + \dots + \text{span}\{v_n\}$.

$$\subseteq) \text{ We } \text{span}\{v_1, \dots, v_n\} \text{ takes the form } w = c_1 v_1 + \dots + c_n v_n \text{ for}$$

$$\text{some } c_i. \text{ Since } c_i v_i \in \text{span}\{v_i\}, w \in \text{span}\{v_1\} + \dots + \text{span}\{v_n\}$$

$\supseteq$) Almost identical.